

Multiple Choice (10 marks)

1. Which of the following sentence is FALSE regarding regression?
 - (a) It relates inputs to outputs.
 - (b) It is used for prediction.
 - (c) It may be used for interpretation.
 - (d) It discovers causal relationships
2. Which among the following prevents overfitting when we perform bagging?
 - (a) The use of sampling with replacement as the sampling technique
 - (b) The use of weak classifiers
 - (c) The use of classification algorithms which are not prone to overfitting
 - (d) The practice of validation performed on every classifier trained
3. Another term for out-of-distribution detection is?
 - (a) anomaly detection
 - (b) one-class detection
 - (c) train-test mismatch robustness
 - (d) background detection
4. Suppose your model is overfitting. Which of the following is NOT a valid way to try and reduce the overfitting?
 - (a) Increase the amount of training data.
 - (b) Improve the optimisation algorithm being used for error minimisation.
 - (c) Decrease the model complexity.
 - (d) Reduce the noise in the training data.
5. If N is the number of instances in the training dataset, nearest neighbors has a classification run time of
 - (a) $O(1)$
 - (b) $O(N)$
 - (c) $O(\log N)$
 - (d) $O(N^2)$

True / False (10 marks)

Answer True or False

6. True or False: Linear hard-margin SVM can only be applied to datasets that are linearly separable without any misclassification.
7. True or False: Bayesians advocate for the use of prior distributions on parameters in a probabilistic model, unlike frequentists.
8. True or False: RoBERTa pretrains on a corpus that is approximately 10 times larger than the corpus used for BERT's pretraining.
9. True or False: The rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ is 1, indicating a single linearly independent row.
10. True or False: Neural networks can utilize a mix of different activation functions within their architecture.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the index of the last occurrence of a given number in a sorted array.
12. Write a python function to find the maximum length of sublist.

End of Paper